Week 1: Data System Overview

Keeyong Han

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- Career Lessons
- Week1: Data System Overview
- Course Intro
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Instructor Bio

Explore

BS/MS in CS
(1989-1995)

Search Engine
(2000-2004)
Startups (including my own)

Samsung Electronics
(1995-2000)

Yahoo
(2004-2011)

Lack of self-confidence, Anxiety
Didn't know where I was heading

Exploit

Sabbatical
(2012)

Angel Investor
(2016-)

Advisor
(2017-)

UpZen, SJSU
(2024-)

Data Domain
(2012-2018)
Polyvore & Udemy

Nonprofits
(2016-)
Simple Steps,
Bayarea Kgroup

Composure, Consistency, Curiosity
Happy with who I am

My Data Career - From 2000 to 2012

- Worked on web search and some vertical searches
- Exposed to machine learning and large scale data processing

WISEnut
Perfect Communication

yahoo!

hadoop

My Data Career - From 2012 to 2018

- Built data teams from scratch in Polyvore (15 ppl) and Udemy (30 ppl)
  a. Polyvore was acquired by Yahoo in 2015
  b. Udemy went IPO in Oct 2021

amazon REDSHIFT

HIVE

Apache Airflow

My Data Career - From 2018 to Now

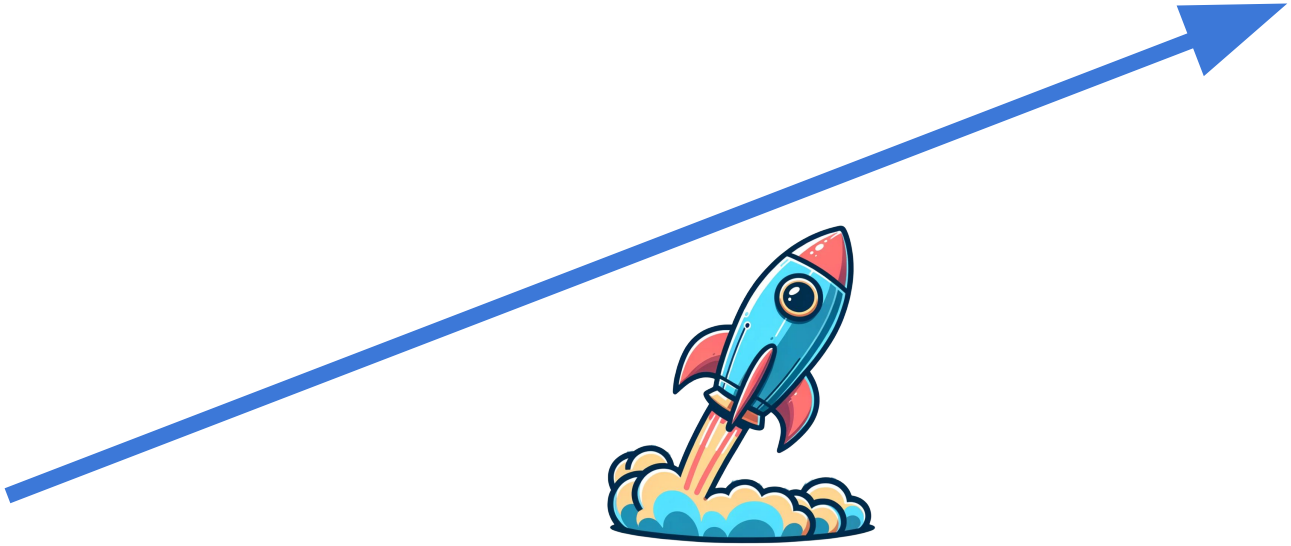
- Consulting/Advising for various companies in data domain
  a. Skip Scooters: Built its data infra on **Google Cloud** (Big Data and IoT)
  b. Blind, Jobox, Tulip, Airvet, …: Built its data infra on **AWS**
  c. Harmonize Health: Built the data infra on **Azure**
  d. SK Telecom: Advised on recommendation engine and AB testing
  e. Hyundai Card: Advised on data team building and recruiting

HARMONIZE
Population Health Simplified

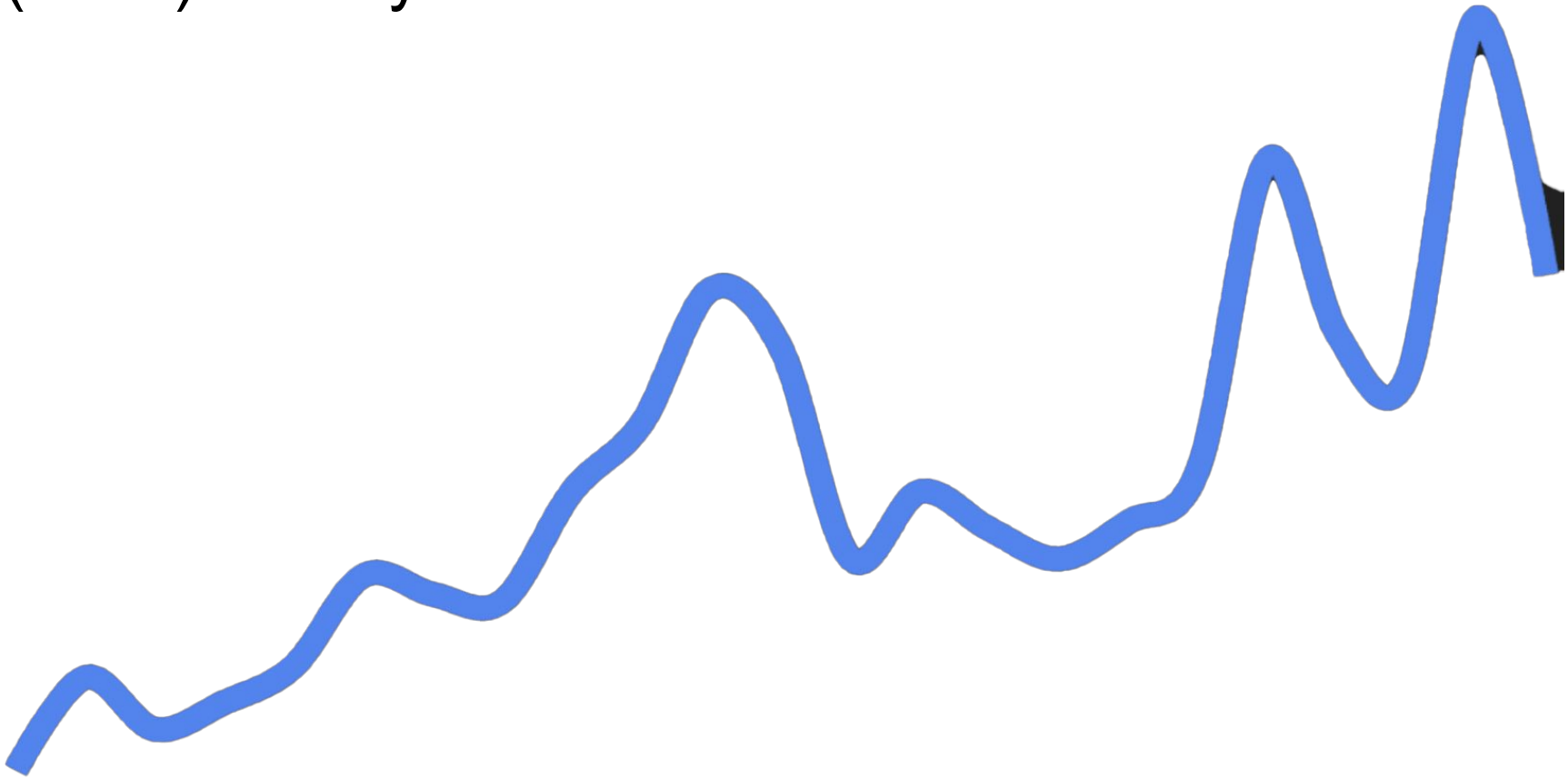
blind

Career Lessons

Expectation

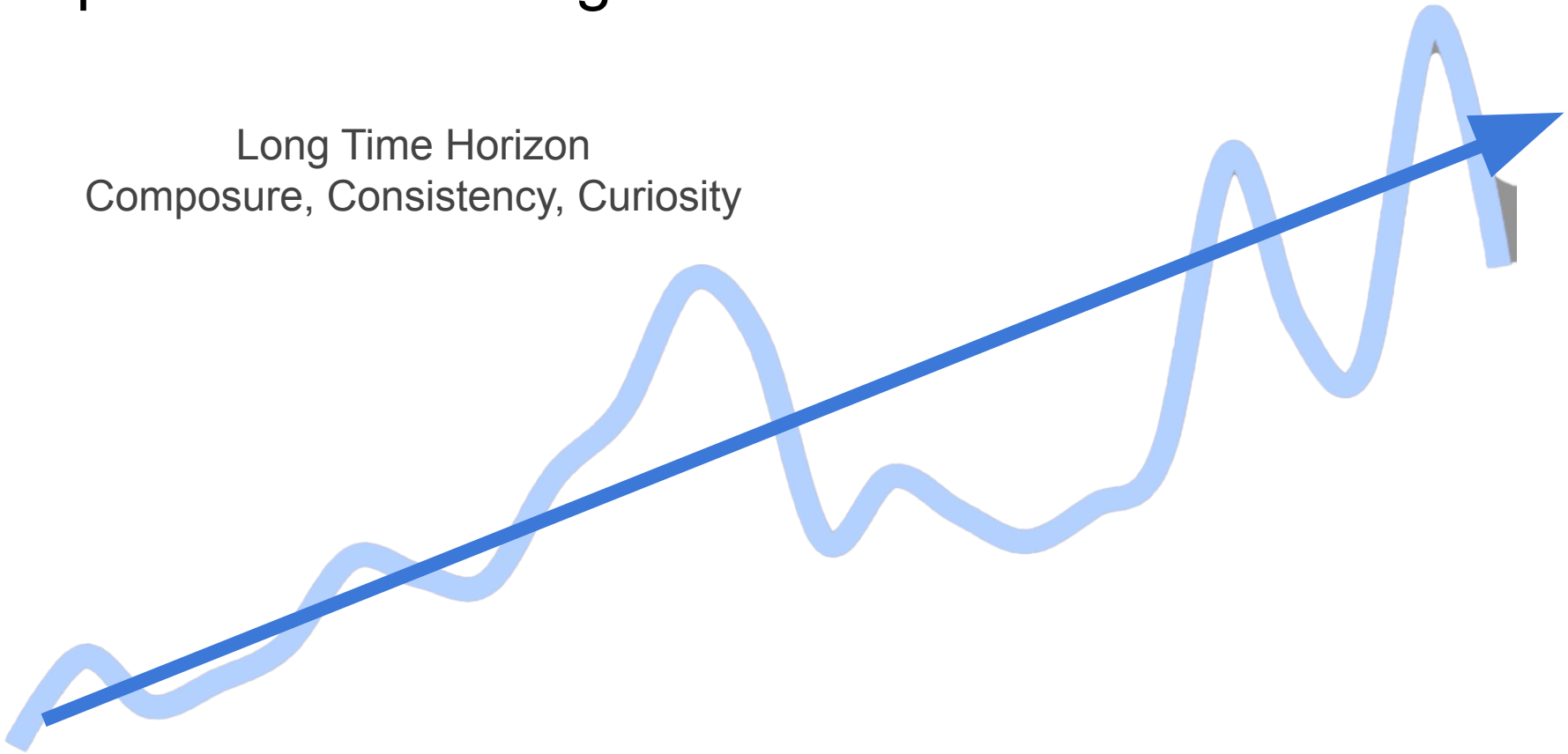


(Ideal) Reality

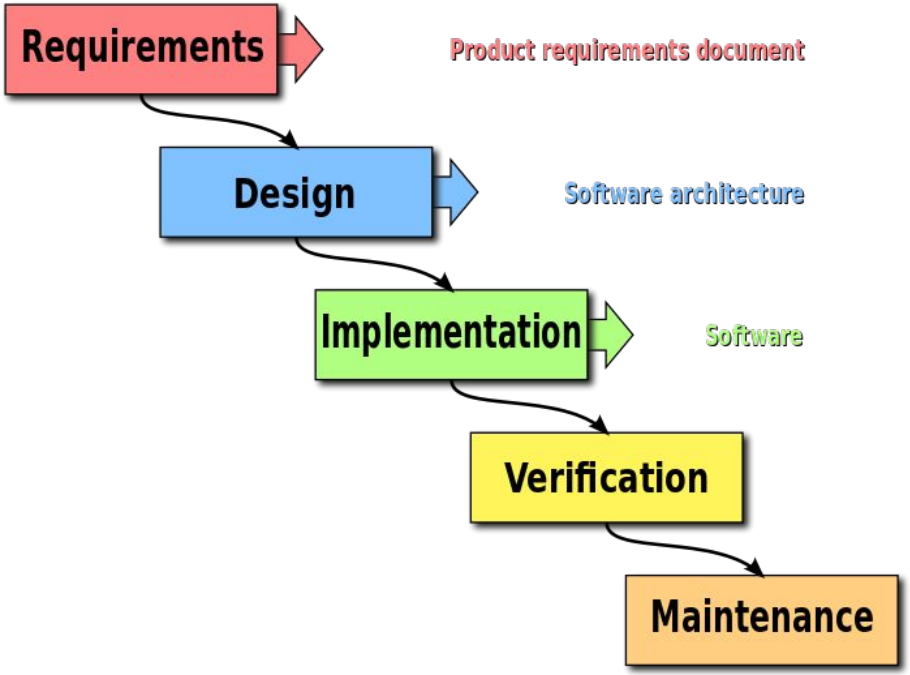


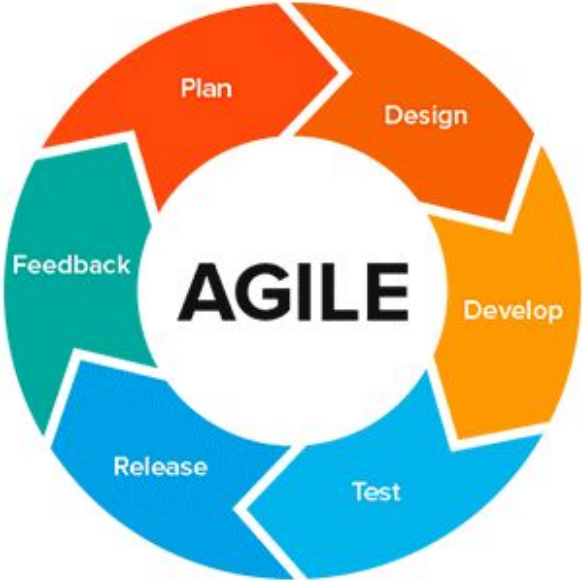
Improve Your Average

Long Time Horizon
Composure, Consistency, Curiosity



Water-Fall vs. Agile





Source: https://en.wikipedia.org/wiki/Waterfall_model

Source: https://www.wearemarketing.com/

Career Lessons

- Life is long; You have plenty of time ahead of you
- Focus on yourself and present (in the right environment)
  a. Do not compare yourself to others
- Have big dreams but live each day fully
  a. Composure & Consistency & Curiosity
  b. Consistency over intensity
- Be result-oriented rather than technology-oriented
  a. Define problems first (Communication skills)
- Don’t be afraid of changes/failures
- Love yourself: Accept yourself as you are

Week 1. Data System Overview

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- Goals of Data Organization
- Participants of Data Organization
- Data Organization Structure
- Lessons Learned

Goals of Data Organization

Data Flow

Site traffic &
3rd party data

1

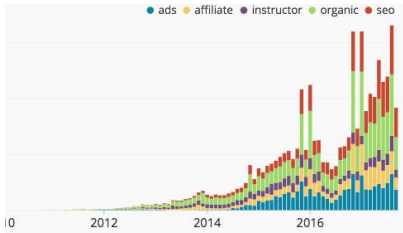
ELT

ETL

Data warehouse

Data Team

2



Business
Insights

3

Site Services

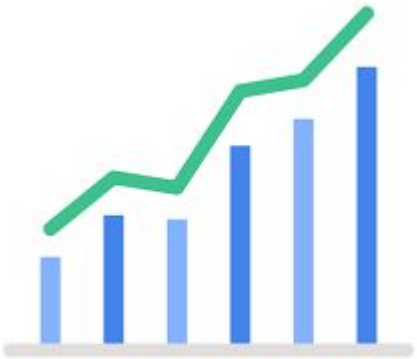
Scalable Product Improvement

Vision of Data Organization

Build **Leverage** for the company through **Trustworthy** Data

How to Add Values: 1. Decision Science

- Equips decision makers for more effective choices
  - **data informed** decisions
    - vs. data driven decisions



How to Add Values: 2. Product Science

- Scalable “product” improvement by algorithm (ML/AI)
  - Ex) Personalized recommendation, search and so on

Data Flow

Site traffic &
3rd party data

1

ELT

ETL

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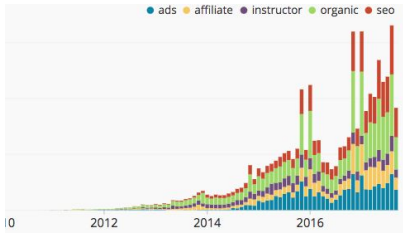


Business
Insights

3

Site Services

Scalable Product Improvement

Participants of Data Organization

Who’s Who in A Data Team

- **Data Engineer**
- Data Analyst
- Data Scientist

Analytics Engineer, ML Engineer, Privacy Engineer, …
MLOps, LLMOps, DataOps, AI Engineer …

Roles of Data Engineers

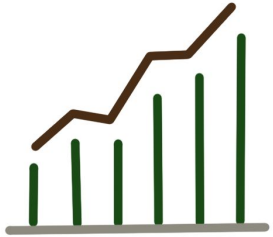
- Building Data Platform/Infra
  - Data Warehouse, Data Lake
  - Data Pipeline (Batch and Streaming)
    - MLOps, LLMOps, …
  - Trend: Central to De-centralized for big organizations
- Supporting Decision/Product Scientists
- Troubleshooting Data Related Issues
  - Data Discovery Issues
  - Data Quality Issues
  - SLA Issues

Data System Evolution

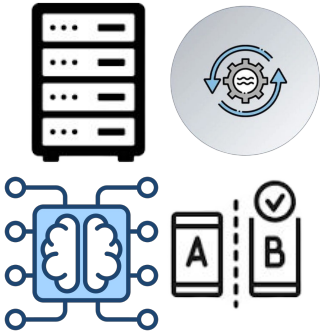
Data Sources

Data Pipelines

Data Storages



BI Tools



Skill Sets of Data Engineers (**Software** Engineer)

SQL, Data Warehouse
Python, Java

GitHub
CI/CD Pipeline

ETL (Airflow), ELT (dbt)
Cloud (AWS,GCP,Azure)

Troubleshooting
Data Literacy
PII Protection (GDPR, CPRA)
Use GenAI tools everyday

- ChatGPT, Claude, …
- Cursor, Gemini CLI, Claude Code, …

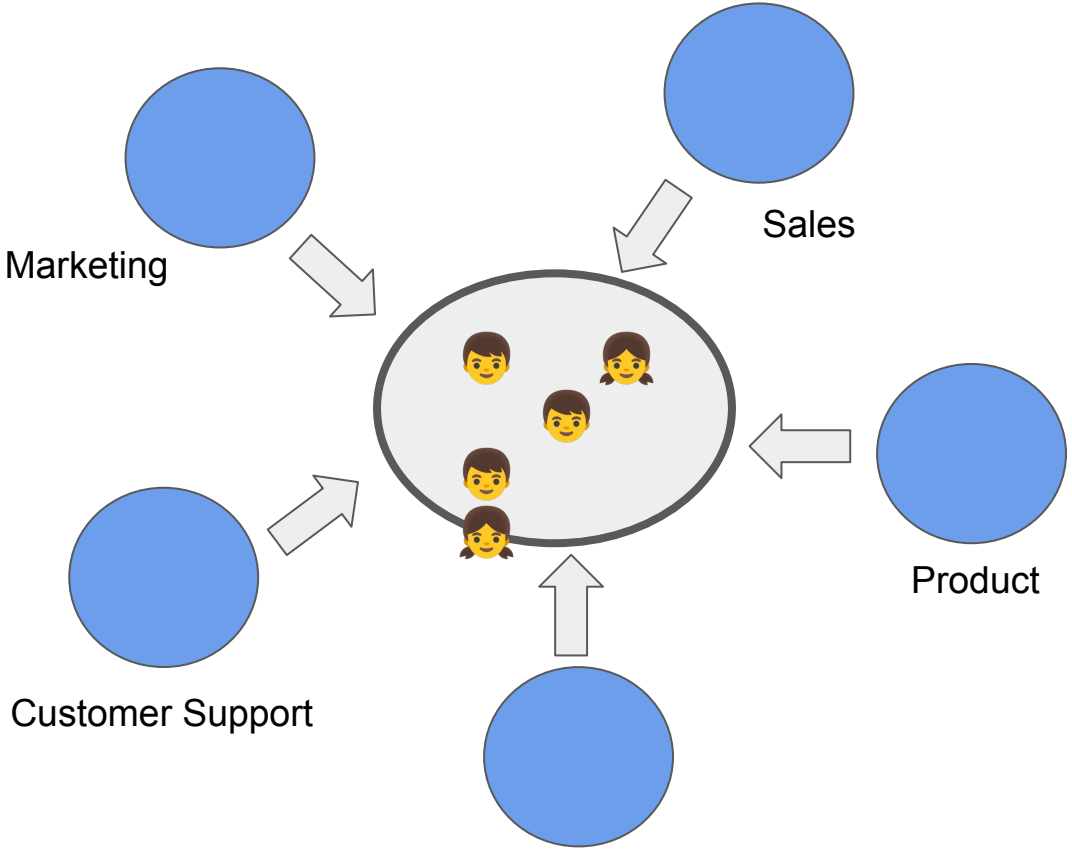
Big Data Processing (Spark/Hadoop)
Streaming Data Processing (Kafka/Flink)

Container Technology (Docker,K8S)

NoSQL (MongoDB, ElasticSearch, Cassandra, DynamoDB …)

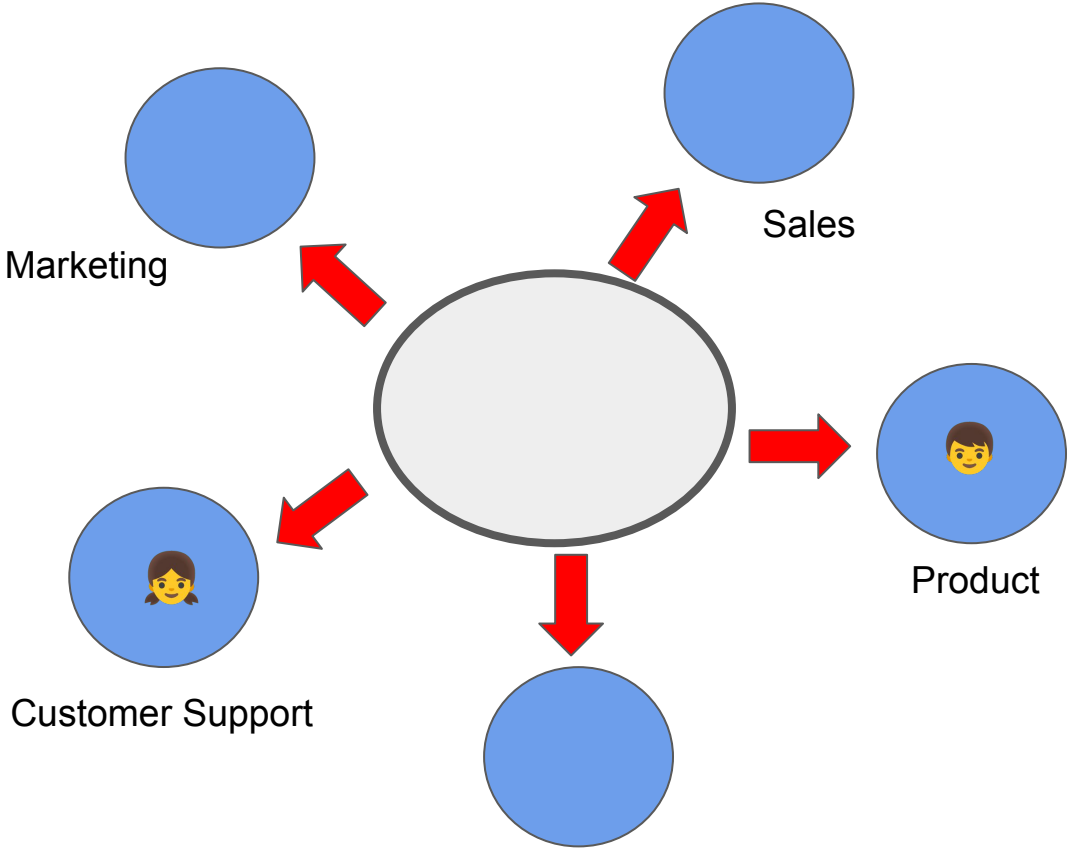
Data Organization Structure

Centralized



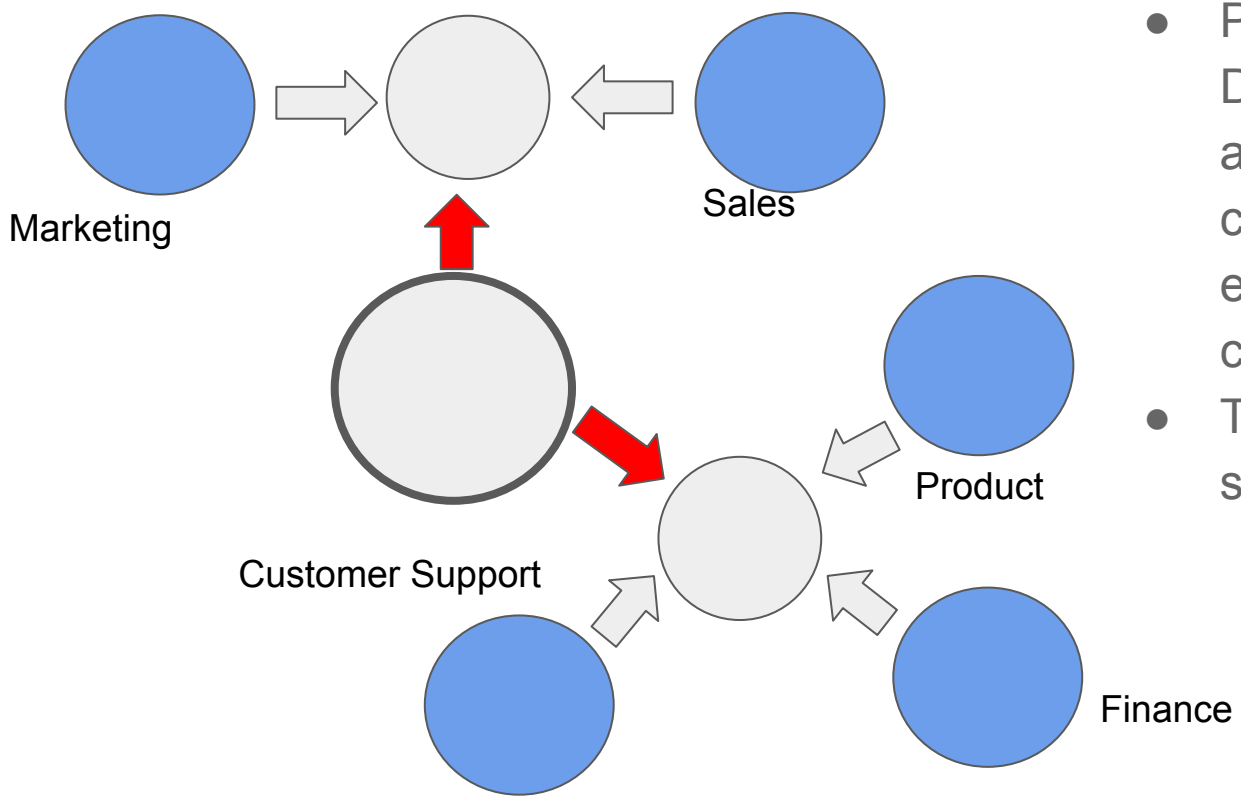
- Priorities and tasks are determined by the central team
- Knowledge sharing and collaboration among the team increase
- Domain knowledge decreases
- Good for Data team members' career

Distributed or Decentralized



- Data Scientist/Analyst are embedded in different units
- Each unit determines priorities/tasks for the data folks
- Knowledge sharing and collaboration among data people decrease
- Business knowledge of data folks improves

Hybrid of Centralized and Distributed



- Priorities and tasks of Data Scientist/Analyst are determined by the central team and each external team collectively
- The most ideal structure
  - Emergence of Citizen Data Scientist/Analyst

Lessons Learned

Data Should Drive Business Results

- Costs related to Data is high:
  - $$$$$$ ⇧
- If no business impact such as revenue increase?
  - Why are they doing? Why do we need them?
  - Measuring the impact objectively and accurately is very critical
- Manage Up and Sideway
  - Many executives heard that data is important but they don’t know why
  - Needs to educate people around the company

Infrastructure (and best practices) First

- Building a scalable infrastructure too early is an overkill
  - Survival is more important for any early stage startup
  - But needs to understand and exercise best practices from day one
- Hire data engineer first with some analytical/ML skill
  - Fullstack Data Professional
- Cloud vs. On-premise
- Batch vs. Real Time

Quality of Data is Important

- Where does a data scientist/analyst spend most time?
- Need to have source of truth for your key data
  - Where is your key data? Can we trust?
  - Avoid vicious cycle of reinvention of similar data
- Requires persistent efforts from the beginning
  - Establish consistent naming scheme for tables and fields
  - Quarterly clean-up offsite

Metrics First

- Every feature implementation need metrics
  - Measure for success or failure
  - You need a hypothesis to develop insights
- Measure metrics objectively is critical
  - A/B Test or Offline test or user survey!
  - This makes automation of manual work easier (fear of AI)
- Data Infra also needs metrics
  - SLA for core pipelines, ...

The Simpler a Solution is The Better

- Start with a simple solution end-to-end quickly
  - Sizing up complexity and impact early on is very important
  - Non-technical blocker is always more complicated
- Rule based solution is better
  - As long as it works!
  - Deep learning is an overkill in many use cases
- Keep “iterations” in mind
  - Going one cycle end-to-end fast is critical

Summer Internship Tips from Students

✅ Communication and understanding what the stakeholder wants is crucial. It’s best to take notes in every meeting so I can refer back to the requirements.
✅ Communicate first, then act in the right direction.
✅ Instead of trying to deliver a perfect project, it’s better to ship a first working version so people can see progress, then refine on the fly.
✅ Asking for help is okay; it’s not seen as weak. But it’s important not to repeat the same mistake again and again.
✅ Explain my analysis/dashboard with clarity and how it will be useful for the stakeholder.
✅ Networking is SUPER IMPORTANT—setting up meaningful coffee chats with recruiters/hiring managers and asking thoughtful product questions really helps.

Course Intro

Goal of the Course

Provide education needed to **start a career as a Data Engineer or ML Engineer**

- Surveyed 10 companies in the Bay Area to see their data tech stack
  - Uber, Airbnb, Pinterest, Apple, Udemy and so on

Prerequisites

- SQL
- Python
- Resources
  - Snowflake
  - Docker in your laptop
  - Google Colab (Python Notebooks)
  - Tableau or Preset or Power BI

First Half Schedule

Week 1	Data System Overview	No homework
Week 2	Data Warehouse Overview	Lab 1 assigned
Week 3	Data Warehouse Deepdive (Snowflake)	
Week 4	Advanced SQLs (on Snowflake)	Quiz 1
Week 5	Data Pipeline Overview	Group Project assigned
Week 6	Airflow Overview	Group Project Team selection due
Week 7	Advanced Airflow	Quiz 2. Lab 1 due
Week 8	Containerization of Airflow	Lab 2 assigned Group Project Proposal due

Second Half Schedule

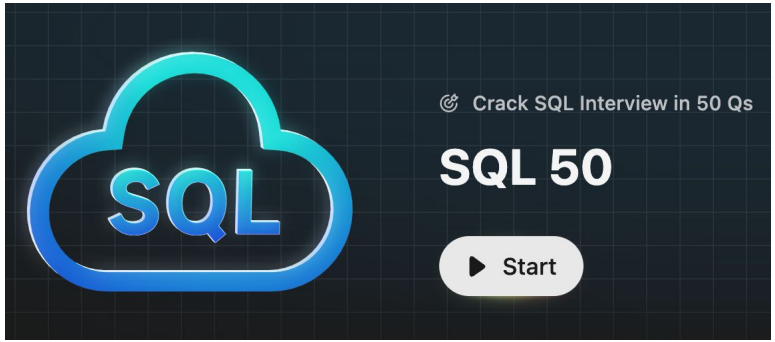
Week 9	ELT Overview	
Week 10	Mid-term	No homework
Week 11	Spring Break	No class
Week 12	ELT Deepdive (dbt)	
Week 13	MLOps Pipeline	
Week 14	Advanced Data Pipelines (LLM)	Quiz 3 Lab 2 due
Week 15	End-to-End Data System	No homework. Guest Speaker
Week 16	Group Project Presentation	No homework. Guest Speaker
Week 17	Final Exam	5:30 to 7:30pm 5/14

Book Recommendations

1. AI Engineering: Building Applications with Foundation Models, Chip Huyen
2. Designing Machine-Learning Systems, Chip Huyen
3. Designing Data-Intensive Applications 2nd edition: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems, Martin Kleppmann

You can read this from oreilly.com for free

https://leetcode.com/studyplan/top-sql-50/



Evaluations

11 Homework assignments (due in a week)	12%
2 Labs (due in a month). 2 students/team	14%
Course Group Project. 4 students/team - Proposal due in week 8 & Presentation in week 16 - Teamwork is a important dimension of the evaluation	18%
3 Quizzes (**best** 2 out of 3)	14%
Midterm Exam (Week 8)	20%
Final Exam (Week 17)	22%

- Quizzes, Midterm and Final will require Respondus LockDown Browser + Webcam in person
- Ensure that the laptop is in a good condition to be able to take exams
- Exams missed or jeopardized due to technical issues with the laptop may not be re-administered
- Final grades will be curved based on overall class performance

Office Hours

- Instructor: Keeyong Han
  - keeyong.han@sjsu.edu
  - 2 to 3PM Thursdays at ISB 863
- ISA: Jeff Chong
  - junjie.chong@sjsu.edu

Introducing Jeff Chong

- 4th semester student
- Office hours: Fridays 3pm to 4pm (Zoom link will be provided on Canvas)

Q&A